§ 4 Oscillator

1. Harmonic Potential

$$V(x) = V(0) + \frac{1}{2}kx^2 + \cdots$$

$$H = T + V = \frac{p^2}{2m} + \frac{1}{2}kx^2 = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

$$= \sum_{ij} \frac{1}{2} p_i (m^{-1})_{ij} p_j + \frac{1}{2} x_i k_{ij} x_j$$

2. Heisenberg Mechanics

$$\ddot{x} + \omega^2 x = 0$$

$$(-\omega_{mn}^2 + \omega^2) x_{mn} = 0 \qquad x = e^{i\omega_{mn} t} x_{mn}$$

$$x_{mn} \neq 0 \iff E_m - E_n = \hbar\omega_{mn} = \pm\hbar\omega$$

$$\hat{\omega} = \begin{pmatrix} 0 & \omega & & \\ -\omega & 0 & \omega & \\ & -\omega & \ddots & \\ & & & \ddots \end{pmatrix}$$

$$2m \sum_k |x_{nk}|^2 \omega_{nk} = \hbar$$

$$2m\omega \sum_n \left(|x_{n,n+1}|^2 - |x_{n,n-1}|^2 \right) = \hbar$$

$$\Rightarrow |x_{n,n+1}|^2 = \frac{n\hbar}{2m\omega}$$

$$E_{nn} = \frac{m}{2}\left(\omega_{n,n+1}^2 |x_{n,n+1}|^2 + \omega_{n,n-1}^2 |x_{n,n-1}|^2 \right)$$

$$= \frac{m\omega^2}{2}\left(\frac{(n+1)\hbar}{2m\omega} + \frac{n\hbar}{2m\omega} \right) = (n+\tfrac{1}{2})\hbar\omega$$

3. Schrodinger Equation

$$-\frac{\hbar^2}{2m}\psi''(x) + \frac{1}{2}m\omega^2 x^2 \psi(x) = E\psi(x)$$

$$\psi''(x) + \left(\frac{2mE}{\hbar^2} - \frac{m^2\omega^2 x^2}{\hbar^2} \right)\psi(x) = 0$$

$$x \to \pm\infty : \quad \psi''(x) - \left(\frac{m\omega x}{\hbar}\right)^2 \psi(x) = 0 \quad , \quad \psi(x) = e^{-\frac{m\omega x^2}{2\hbar}}$$

Let $\psi(x) = e^{-\frac{m\omega x^2}{2\hbar}} u(x)$

$$\left[\left(-\frac{m\omega x}{\hbar}\right) \psi(x) + e^{-\frac{m\omega x^2}{2\hbar}} u'(x) \right]' + \left[\frac{2mE}{\hbar^2} - \left(\frac{m\omega x}{\hbar}\right)^2 \right] \psi(x)$$

$$= -\frac{m\omega}{\hbar} e^{-\frac{m\omega x^2}{2\hbar}} u(x) - \frac{m\omega x}{\hbar} e^{-\frac{m\omega x^2}{2\hbar}} u'(x) + e^{-\frac{m\omega x^2}{2\hbar}} u''(x) + \frac{m^2\omega^2 x^2}{\hbar^2} e^{-\frac{m\omega x^2}{2\hbar}} u(x) = 0$$

$$h''(x) - 2\frac{m\omega x}{\hbar} h'(x) + \left[\frac{2E}{\hbar\omega} - \frac{m\omega}{\hbar}\right] h(x) = 0$$

$$h(x) = \sum_n a_n x^n$$

$$(n+2)(n+1)\, a_{n+2} - 2\frac{m\omega}{\hbar} n\, a_n + \left(\frac{2E}{\hbar\omega} - \frac{m\omega}{\hbar}\right) a_n = 0$$

$$a_{n+2} = \frac{1}{(n+2)(n+1)} \frac{m\omega}{\hbar} \left\{ \frac{2E}{\hbar\omega} - m\omega(1+2n) \right\} a_n$$

$\exists\, n$ s.t. $(2n+1)\hbar\omega - \frac{2E}{m\omega} = 0 \quad \Rightarrow \quad E = (n+\frac{1}{2})\hbar\omega$

⇒ ① 偶数 $a_{2n+1} = 0$, $a_1 = 0$

② 奇数 $a_{2n+1} \neq 0$, $a_0 = 0$

$N = 0$: $a_2 = 0$, $a_1 = 0$ $\quad \psi_0(x) \sim e^{-\frac{m\omega x^2}{2\hbar}}$

$N = 1$: $a_3 = 0$, $a_0 = 0$ $\quad \psi_1(x) \sim x e^{-\frac{m\omega x^2}{2\hbar}}$

$N = 2$: $a_4 \neq 0$, $a_1 = 0$

$$a_2 = \frac{1}{1\cdot 2}\frac{m\omega}{\hbar}\left(\hbar\omega - 2\cdot(2+\tfrac{1}{2})\hbar\omega\right) a_0 = -2\frac{m\omega}{\hbar} a_0$$

$$\psi(x) \sim \left(1 - \frac{2m\omega x^2}{\hbar}\right) e^{-\frac{m\omega x^2}{2\hbar}}$$

4. Dirac Operator.

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2 = \left(\frac{p}{\sqrt{2m}} + i\sqrt{\frac{m}{2}}\omega x\right)\left(\frac{p}{\sqrt{2m}} - i\sqrt{\frac{m}{2}}\omega x\right) + \frac{i}{2m}[x,p]$$

$$= \hbar\omega\left(\frac{p}{\sqrt{2m\hbar\omega}} + i\sqrt{\frac{m\omega}{2\hbar}}x\right)\left(\frac{p}{\sqrt{2m\hbar\omega}} - i\sqrt{\frac{m\omega}{2\hbar}}x\right) + \frac{1}{2}\hbar\omega$$

$$= \hbar\omega\, a^\dagger a + \frac{1}{2}\hbar\omega$$

$$a^\dagger = \frac{p}{\sqrt{2m\hbar\omega}} + i\sqrt{\frac{m\omega}{2\hbar}}x \;, \quad a = \frac{p}{\sqrt{2m\hbar\omega}} - i\sqrt{\frac{m\omega}{2\hbar}}x$$

$$[a, a^\dagger] = \frac{1}{2m\hbar\omega} \cdot i\sqrt{\frac{m\omega}{2\hbar}}\left([x,p] - [p,x]\right)$$

$$= \frac{1}{2m\hbar\omega} \cdot \sqrt{\frac{m\omega}{2\hbar}}\, 2i\hbar = 1$$

$\vec{x}$ 数算子 $N = a^\dagger a$, $N|n\rangle = n|n\rangle$

$$[a^\dagger, N] = -a^\dagger, \quad [a, N] = a$$

$$\begin{cases} (a^\dagger N - N a^\dagger)|n\rangle = -a^\dagger|n\rangle, & N a^\dagger|n\rangle = a^\dagger(1+N)|n\rangle = (1+n)a^\dagger|n\rangle \\ (aN - Na)|n\rangle = a|n\rangle, & N a|n\rangle = (n-1)a|n\rangle \end{cases}$$

$a^\dagger$：生算符，　a：湮灭算符。

5. 粒子数表象：

$$\langle n|N|n\rangle = \langle n|a^\dagger a|n\rangle = n$$

$$\begin{cases} a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle \\ a|n\rangle = \sqrt{n}\,|n-1\rangle \end{cases}$$

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle$$

$$\begin{cases} p = \sqrt{\frac{m\hbar\omega}{2}}(a^\dagger + a) \\ x = \frac{1}{i}\sqrt{\frac{\hbar}{2m\omega}}(a^\dagger - a) \end{cases}$$

6. 在坐标表象下：

$a|0\rangle = 0$　　$|a|0\rangle|^2 = \langle 0|a^\dagger a|0\rangle = \langle 0|N|0\rangle = 0$

$$\langle x|a|0\rangle = \langle x|\frac{p}{\sqrt{2m\hbar\omega}}|0\rangle - \langle x|i\sqrt{\frac{m\omega}{2\hbar}}x|0\rangle$$

$$= \frac{1}{\sqrt{2m\hbar\omega}}\langle x|p|0\rangle - i\sqrt{\frac{m\omega}{2\hbar}}\langle x|x|0\rangle$$

$$= \frac{1}{\sqrt{2m\hbar\omega}}(-i\hbar)\frac{d}{dx}\psi_0(x) - i\sqrt{\frac{m\omega}{2\hbar}}x\,\psi_0(x)$$

$$= 0$$

$$\Rightarrow \psi_0'(x) - \frac{m\omega}{\hbar}\psi_0(x) = 0, \quad \psi_0(x) = e^{-\frac{m\omega x^2}{2\hbar}}$$

$$\psi_n(x) = \langle x|n\rangle = \langle x|\frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle$$

$$\sim \left(\frac{d}{dx} - \frac{m\omega}{\hbar}x\right)^n \psi_0(x)$$

$$\propto H_n\left(\sqrt{\frac{m\omega}{2\hbar}}x\right)$$

Hermite多项式：H_n 有 n 个零点。

$\Rightarrow \psi_n(x)$ 有 n 个零点。

7. 不确定关系

$$\langle n|(\Delta x)^2|n\rangle = \langle n|x^2|n\rangle - \langle n|x|n\rangle^2$$

$$= -\frac{\hbar}{2m\omega}\left\{\langle n|(a-a^\dagger)^2|n\rangle - \left(\langle n|a-a^\dagger|n\rangle\right)^2\right\}$$

$$\langle n|(\Delta p)^2|n\rangle = \langle n|p^2|n\rangle - \langle n|p|n\rangle^2$$

$$= \frac{1}{2}m\hbar\omega\langle n|(a+a^\dagger)^2|n\rangle$$

$$\langle(\Delta x)^2\rangle_n\langle(\Delta p)^2\rangle_n = \langle n|\Delta x^2|n\rangle\langle n|(\Delta p)^2|n\rangle$$

$$= -\frac{1}{4}\hbar^2\langle n|-(a^2a+aa^\dagger)|n\rangle\langle n|(a^\dagger a+aa^\dagger)|n\rangle$$

$$= \frac{1}{4}\hbar^2\langle n|a^\dagger a+aa^\dagger|n\rangle^2$$

$$= \frac{1}{4}(2n+1)^2\hbar^2 = \left[\left(n+\frac{1}{2}\right)\hbar\right]^2$$

$|n\rangle$ 在 $(n\to\infty)$ 不对应经典

8. 含时演化

Heisenberg Picture:

① 证明 $a_H = [a_H, H] = [a_H, \hbar\omega(a_H^\dagger a_H + \frac{1}{2})]$

$$= \hbar\omega[a_H, a_H^\dagger a_H] = \hbar\omega\{[a_H, a_H^\dagger]a_H + a_H^\dagger[a_H, a_H]\}$$

$$= \hbar\omega a_H$$

$$\to a_H(t) = a_H(0)e^{-i\omega t}$$

② $a(t) = e^{iHt/\hbar}a(0)e^{-iHt/\hbar}$

$e^ABe^{-A} = B + [A,B] + \frac{1}{2!}[A,[A,B]] + \frac{1}{3!}[A,\dots]$

$$= a(0) + \frac{it}{\hbar}[H, a(0)] + \frac{1}{2!}\left(\frac{it}{\hbar}\right)^2[H,[H,a(0)]] + \cdots$$

$$= a(0) + \frac{it}{\hbar}\hbar\omega[(a^\dagger a+\tfrac{1}{2}), a] + \frac{1}{2!}(i\omega t)^2[(a^\dagger a+\tfrac{1}{2}),[a^\dagger a+\tfrac{1}{2}, a]]$$

$$= a(0)\left(1 + (-i\omega t) + \frac{1}{2!}(-i\omega t)^2 + \cdots\right)$$

$$= a(0)e^{-i\omega t}$$

$$X(t) = i\sqrt{\frac{\hbar}{2m\omega}}\left(a(0)e^{-i\omega t} - a^\dagger(0)e^{i\omega t}\right)$$

$$= \sqrt{\frac{m\hbar\omega}{2}}\left(\cdots\right)$$

9. 相干态 (一系列 n 最小 / 相干程度的态)

Def: $a|\alpha\rangle = \alpha|\alpha\rangle$

$$|\alpha\rangle = \sum_n c_n|n\rangle$$

$$\Rightarrow a|\alpha\rangle = \alpha|\alpha\rangle = \alpha\sum_n c_n|n\rangle$$

$$= a\sum_n c_n|n\rangle = \sum_n c_n\sqrt{n}|n-1\rangle = \sum_n c_{n+1}\sqrt{n+1}|n\rangle$$

$$\Rightarrow c_{n+1} = \frac{\alpha}{\sqrt{n+1}}c_n, \quad c_n = \frac{\alpha^n}{\sqrt{n!}}c_0$$

注: 归一化: $\sum_n |c_n|^2 = c_0^2 e^{|\alpha|^2} = 1 \Rightarrow c_0 = e^{-\frac{|\alpha|^2}{2}}$

$$|\alpha\rangle = \sum_n e^{-\frac{|\alpha|^2}{2}}\frac{\alpha^n}{\sqrt{n!}}|n\rangle \qquad |n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle$$

$$= e^{-\frac{|\alpha|^2}{2}}\sum_n \frac{(\alpha a^\dagger)^n}{n!}|0\rangle = e^{-\frac{|\alpha|^2}{2}}e^{\alpha a^\dagger}|0\rangle$$

时间演化:

$$|\psi(t)\rangle = e^{-i\omega(a^\dagger a+\frac{1}{2})t}|\psi(0)\rangle$$

$$= e^{-i\omega(a^\dagger a+\frac{1}{2})t}\sum_n \frac{\alpha_0^n}{\sqrt{n!}}e^{-\frac{|\alpha_0|^2}{2}}|n\rangle$$

$$= e^{-\frac{i\omega t}{2}}e^{-\frac{|\alpha_0|^2}{2}}\sum_n \frac{\alpha_0^n}{\sqrt{n!}}e^{-in\omega t}|n\rangle$$

$$= e^{-\frac{i\omega t}{2}}e^{-\frac{|\alpha_0|^2}{2}}\sum_n \frac{(\alpha_0 e^{-i\omega t}a^\dagger)^n}{n!}|0\rangle$$

$$= e^{-\frac{i\omega t}{2}}e^{-\frac{|\alpha_0|^2}{2}}e^{\alpha_0 e^{-i\omega t}a^\dagger}|0\rangle$$

$$= e^{-\frac{i\omega t}{2}}e^{-\frac{|\alpha_0|^2}{2}}\sum_n \frac{(\alpha_0 e^{-i\omega t})^n}{\sqrt{n!}}|n\rangle$$

$$= e^{-\frac{i\omega t}{2}}|\alpha_0 e^{-i\omega t}\rangle$$

相干态演化仍然是相干态.

$$\langle X(t)\rangle_\alpha = \langle\alpha|X|\alpha\rangle = i\sqrt{\frac{\hbar}{2m\omega}}\left(\langle\alpha|a|\alpha\rangle - \langle\alpha|a^\dagger|\alpha\rangle\right)$$

$$= i\sqrt{\frac{\hbar}{2m\omega}}\left(\alpha e^{-i\omega t} - \alpha^* e^{i\omega t}\right) \qquad \alpha = |\alpha| e^{i\phi}$$

$$= |\alpha|\sqrt{\frac{2\hbar}{m\omega}}\sin(\omega t - \phi)$$

在坐标表象下：

$$\langle X|a|\alpha\rangle = \langle X|\alpha|\alpha\rangle = \alpha\langle X|\alpha\rangle$$

$$\frac{(-i\hbar)}{\sqrt{2m\hbar\omega}}\ \psi_\alpha'(x) + i\sqrt{\frac{m\omega}{2\hbar}}\,x\,\psi_\alpha(x) = \alpha\psi_\alpha(x)$$

$$\Rightarrow \quad \psi_\alpha'(x) - \left(\frac{m\omega}{\hbar}x + i\sqrt{\frac{2m\omega}{\hbar}}\alpha\right)\psi_\alpha(x) = 0$$

$$\psi_\alpha'(x) - \frac{m\omega}{\hbar}\left(x - i\sqrt{\frac{\hbar}{m\omega}}\alpha\right)\psi_\alpha(x) = 0$$

$$\psi_\alpha(x) = C e^{-\frac{m\omega}{\hbar}\left(x - i\sqrt{\frac{\hbar}{m\omega}}\alpha\right)^2} = \psi_0\left(x - i\sqrt{\frac{\hbar}{m\omega}}\alpha\right)$$

平移 ~ 对应波包.

作功相关：